Deep Food Image Classifier with CNNs and Evolutionary Hyperparameter Optimization

V Harsha Vardhan Yellela, MSCS Candidate — CJ Chung, PhD, Professor
College of Arts & Science, Lawrence Technological University

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Abstract

Food images often look similar, making classification difficult on small datasets. We evaluate a residual CNN trained on a 10-class food dataset and explore evolutionary hyperparameter optimization methods. Evolutionary search—including ES(1+1), DEAP (μ=5, λ=10), CMA-ES—and random search are compared against a Hyperband-tuned baseline. These strategies substantially boost validation and test accuracies compared to the untuned CNN, while maintaining moderate training times.

Introduction

Classifying food photographs into categories such as *churros*, *french fries* and *waffles* is challenging due to subtle visual cues, varied lighting and limited labelled samples. Deep CNNs extract hierarchical features but require careful hyperparameter tuning. Traditional manual tuning is time-consuming; random search may miss promising regions.

Dataset: The simplified dataset contains 10 food classes: chicken wings, churros, french fries, hamburger, hot dog, ice cream, macaroni and cheese, pancakes, pizza, and waffles. We split 70% of the images for training, 15% for validation and 15% for testing. Data augmentation applies horizontal flips, rotations up to 10°, random zoom and brightness adjustments to boost generalization on small data.

Motivation: Hyperparameters such as learning rate, dropout, label smoothing and augmentation strength have a large impact on CNN performance. We aim to automate this search via evolutionary algorithms.

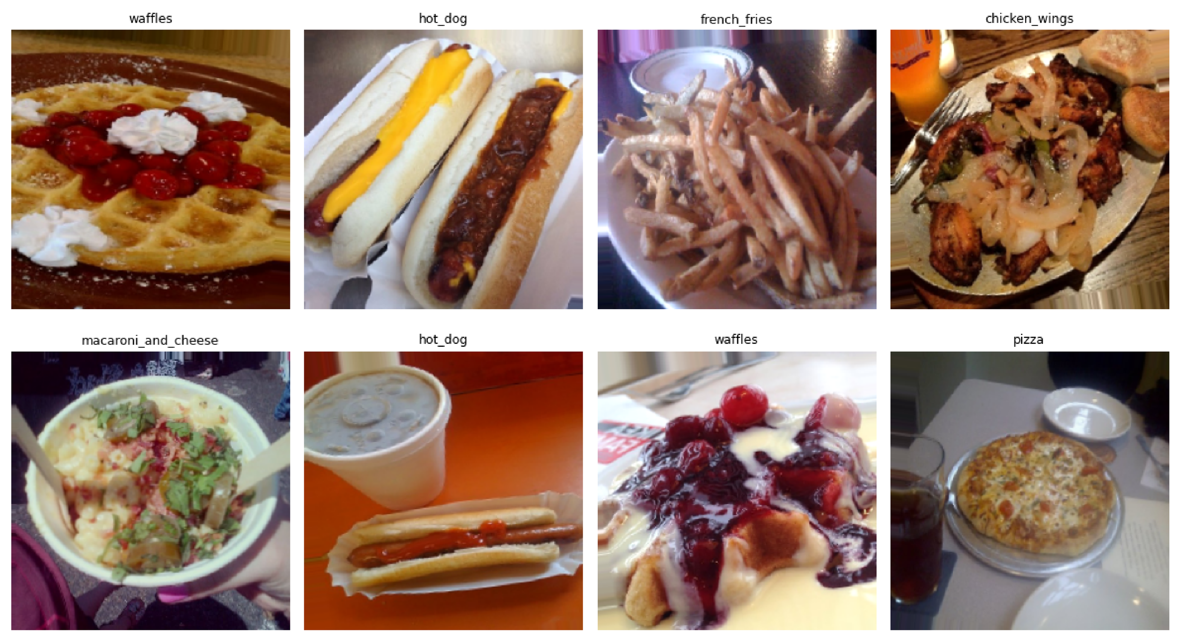

Figure 1: Samples from the simplified 10-class food dataset

System Design & Workflow

The data flow includes image loading, preprocessing and CNN training, followed by evolutionary hyperparameter optimization. All experiments use mixed-precision training on dual GPUs.

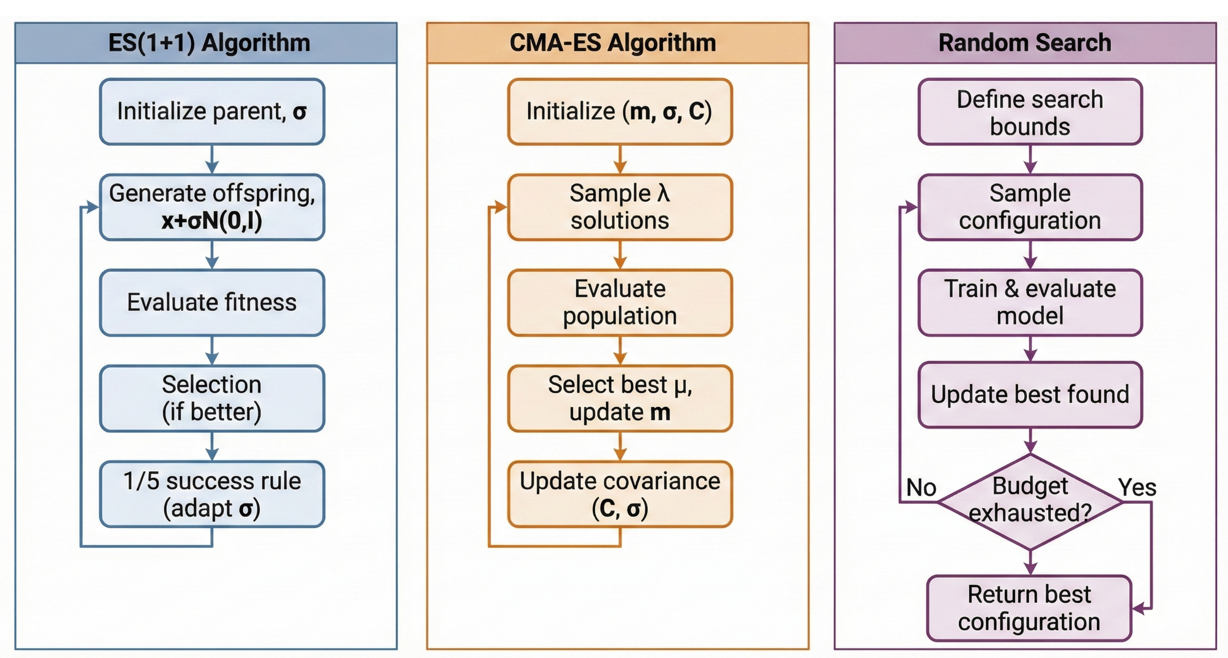

Figure 2: Workflow: input images are resized, normalized and augmented; a ResNet-style CNN predicts class probabilities; a confidence check triggers out-of-scope detection.

Methods

Model Architecture: We adopt a ResNet-style CNN with four residual blocks, expanding channel depth from 64 to 1024, followed by global average pooling and a dense head for ten classes.

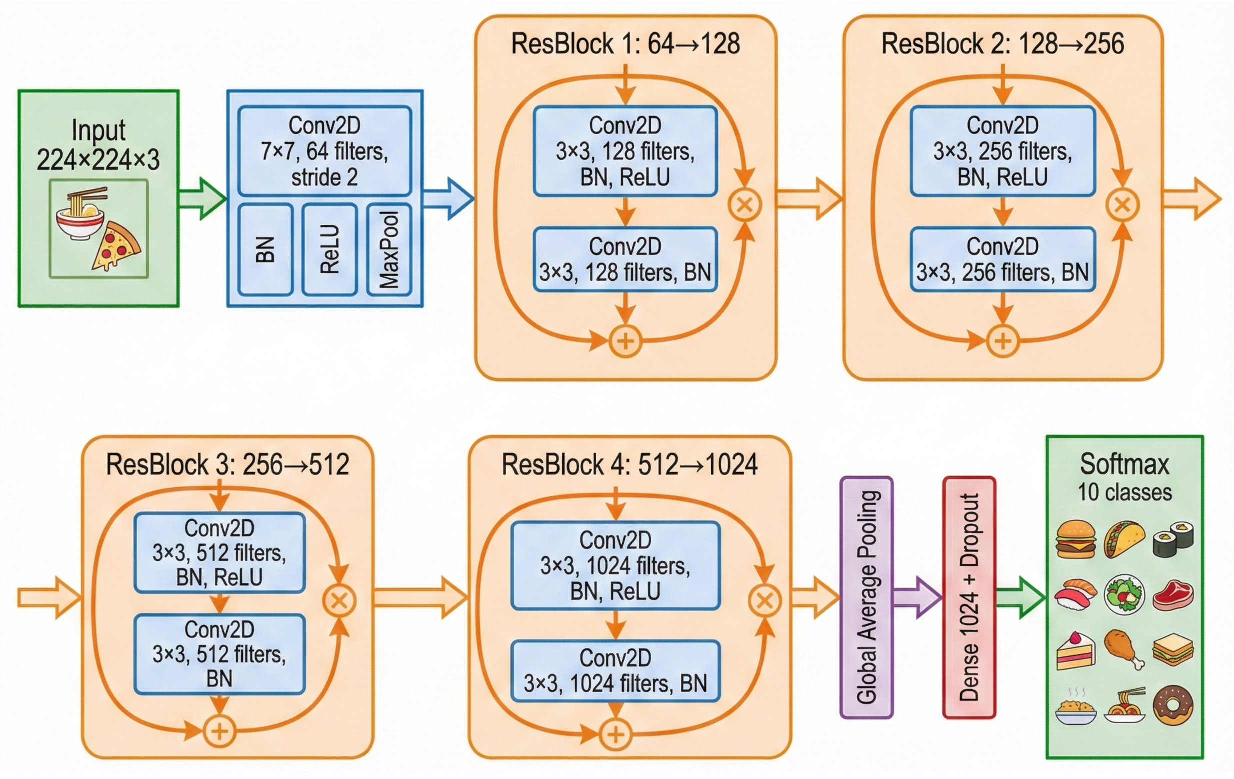

Figure 3: ResNet-style CNN used in all experiments

Optimization Methods:

- **ES(1+1):** Single-parent evolution with Gaussian mutation; mutation variance adapted via the 1/5th success rule.
- **DEAP** (μ=5, λ=10)**:** Population-based evolution (five parents, ten offspring) selecting the best individuals each generation.
- **CMA-ES:** Covariance-matrix adaptation to learn correlations among hyperparameters.
- **Random Search:** Uniform sampling across hyperparameter ranges.
- **Keras Tuner (Hyperband):** Bandit-based search allocating more resources to promising configurations.

Training: All models are trained for up to 50 epochs with early stopping based on validation accuracy. We use cross-entropy loss with label smoothing and Adam optimizer. Data normalization rescales pixel intensities to $[0, 1]$.

Performance Summary:

Method	Val Acc (%)	Test Acc (%)	Δ vs Baseline	Macro F1
Baseline CNN	72.33	72.47	–	0.72
ES(1+1) [Winner]	75.13	79.80	+7.33%	0.80
CMA-ES	77.07	77.13	+4.66%	0.77
Random Search	77.00	77.93	+5.46%	0.78
Keras Tuner	77.47	79.27	+6.80%	0.79
DEAP (μ=5, λ=10)	77.33	79.33	+6.86%	0.79

Table 1: Performance summary of all methods (higher is better).

Key Observations: The baseline CNN underfits due to fixed hyperparameters. ES(1+1) improves accuracy by adapting learning rate, dropout and augmentation ranges over generations. CMA-ES and DEAP both explore a larger hyperparameter space, with DEAP showing stable average improvements across populations. Random search occasionally finds good configurations but lacks directed convergence. Hyperband balances exploration and exploitation by early-stopping poor trials and fully training promising configurations.

Optimization Behavior

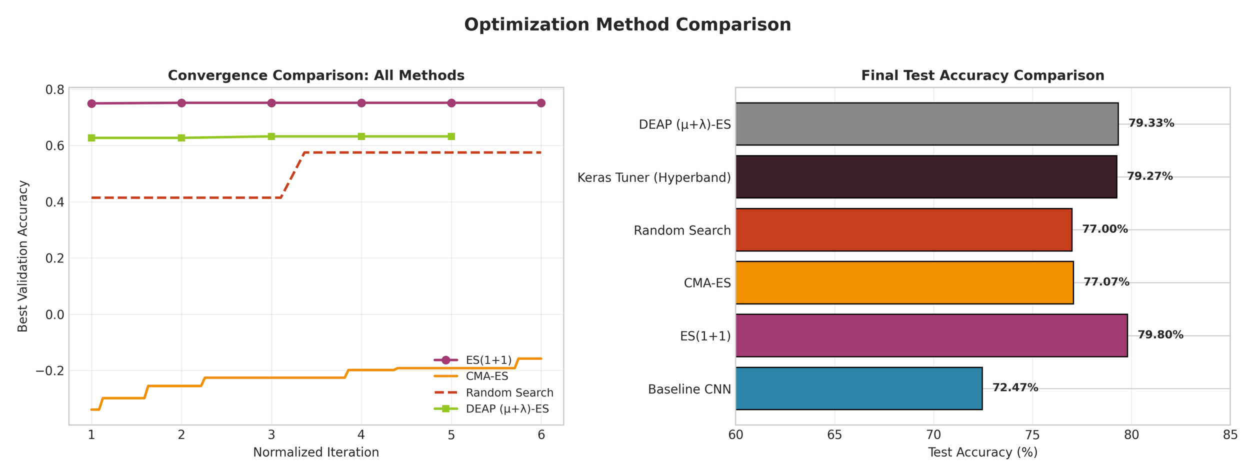

Figure 4: Combined convergence of ES(1+1), DEAP (μ=5, λ=10), CMA-ES and Random Search. Each curve shows best-so-far validation accuracy.

Evolutionary methods steadily improve validation accuracy. ES(1+1) converges quickly and achieves the highest performance. DEAP exhibits a gradual but stable increase. CMA-ES explores widely with slower gains. Random Search shows noisy performance with occasional improvements.

Hyperparameter Analysis

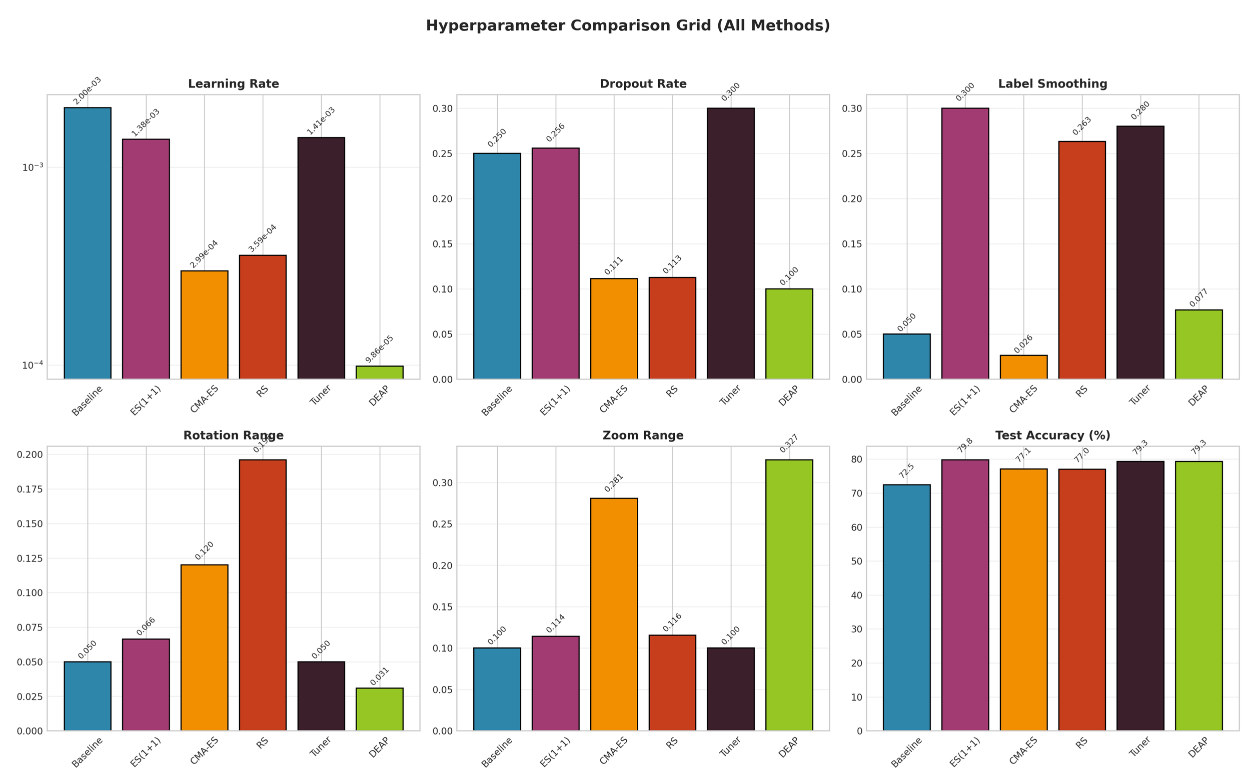

Figure 5: Hyperparameter comparison across optimization methods. Each bar shows the best value of learning rate, dropout, label smoothing, rotation and zoom found by the search.

ES(1+1) and Hyperband prefer moderate learning rates and dropouts; DEAP pushes rotation and zoom augmentation further, indicating the importance of aggressive augmentation on small datasets. CMA-ES yields balanced parameters but lower overall accuracy.

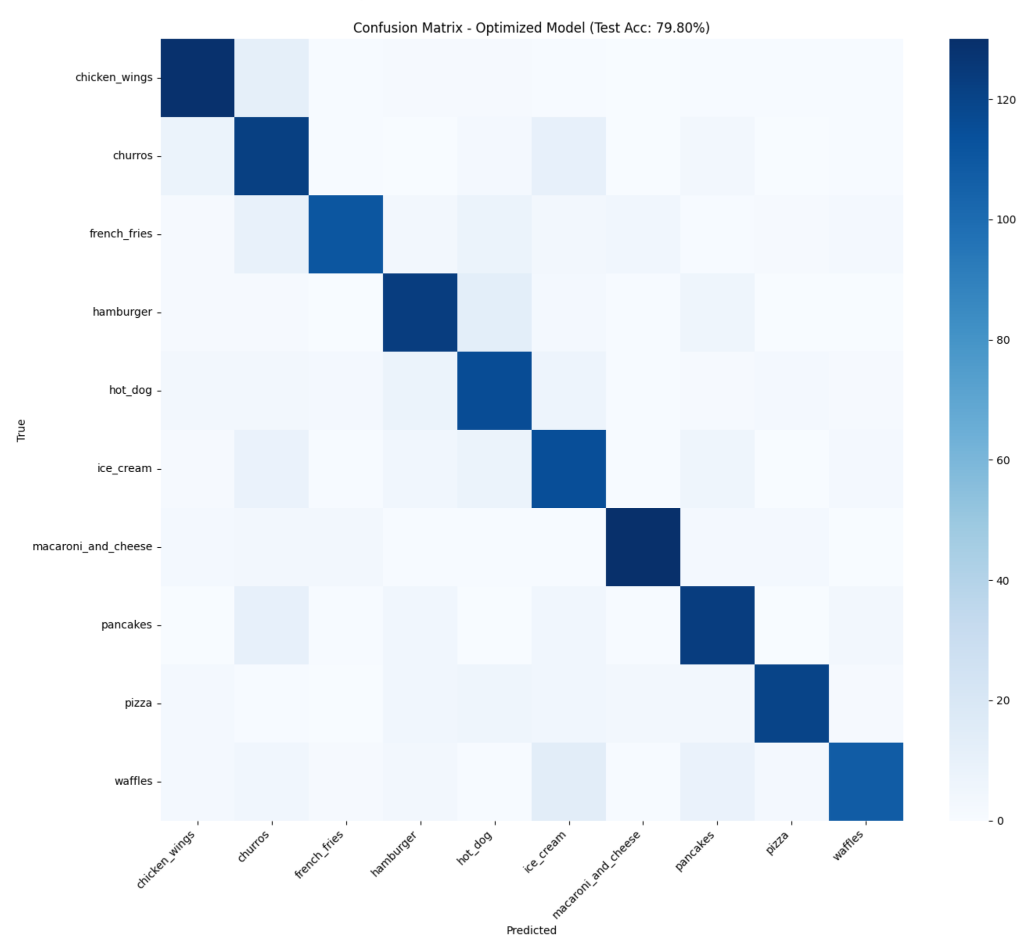

Figure 6: Confusion matrix for the ES(1+1) optimized model on the test set, showing class-wise prediction accuracy. ES(1+1) achieves the best performance at 79.8%.

Evolutionary hyperparameter optimization consistently improves CNN performance over the untuned baseline. ES(1+1) achieves the highest test accuracy (79.8%), followed closely by DEAP (79.3%) and Keras Tuner (79.27%). Population-based methods exhibit more stable validation behavior, while ES(1+1) converges faster with fewer evaluations. Detailed numerical comparisons are reported in Table 1, and convergence behavior is illustrated in Figure4.

Training & Inference

To demonstrate real-world deployment behavior, we visualize inference on an unseen food image. The model outputs softmax probabilities for all classes and reports the top-5 predictions. A confidence threshold of 60

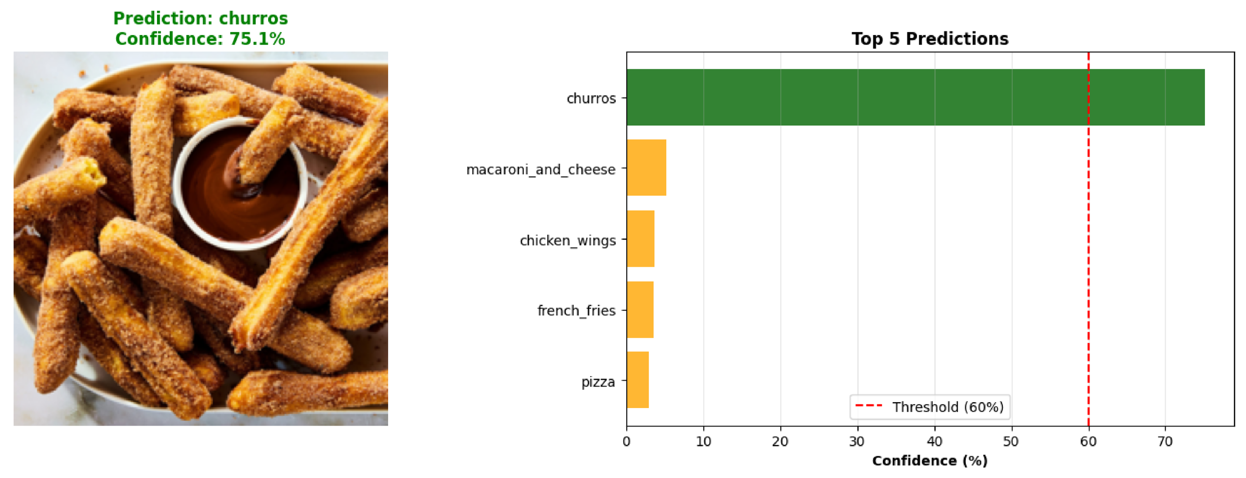

Figure 7: Inference output of the optimized CNN predicting churros (75.1% confidence) with top-5 softmax probabilities and a 60% validation threshold

During inference, the trained model processes a single input image and outputs class probabilities using a softmax layer. The top-5 predictions are displayed along with confidence scores, and a confidence threshold is applied to distinguish valid food predictions from out-of-scope inputs.

Conclusion

Evolutionary hyperparameter optimization significantly enhances CNN performance on small food classification datasets. ES(1+1) provides the highest test accuracy and converges quickly, while DEAP offers robust improvements with population diversity. Random search and CMA-ES lag behind but still outperform the untuned baseline. Hyperband achieves competitive results but requires longer search time. The best models achieve nearly 80% accuracy, a gain of over 7 percentage points relative to the baseline.

Future Work

Future directions include expanding the dataset to more classes, exploring transfer learning with pre-trained CNN backbones, evaluating multi-objective evolutionary strategies (accuracy and latency), and deploying models on embedded devices with quantised weights for real-time inference.

References

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